DGP-by-Estimator Monte Carlo

Methods, corrected results, and interpretation

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Abstract

This simulation applies four procedures—KSS, BLM, Borovičková–Shimer (BS20), and the current project low-rank plug-in—to panels generated from five different worker–firm wage models. The project leave-one-out correction is still under development and is not simulated. The central result is that three distinct issues must be kept separate: sampling error, disagreement between population estimands, and whether an estimator's required support is present. A forensic audit showed that the original BLM summaries mixed supported fits with fits missing stayer-class support. The corrected pipeline rejects such samples before estimation. Of 500 BLM attempts, 243 have complete support, 242 pass the likelihood diagnostic, one returns with a numerical warning, and 257 are classified as unsupported. No supported attempt fails in PyTwoWay. All results below use this corrected classification.

1 Question and scope

The experiment asks one symmetric question: what happens when every procedure is used under every DGP? This is more informative than generating all panels from an additive AKM model because an additive design mechanically favors an additive estimator. The five DGPs are additive AKM, Crippa/Tukey nonadditivity, discrete BLM types, continuous low-rank factors, and GKLP comparative advantage. Each DGP has 100 independent Monte Carlo replications.

The project procedure in this document is the current BIC-selected low-rank plug-in. It is not called the final project LOO estimator because the full LOO correction has not yet been implemented.

Every reported number is calculated from the support-audited merged output. Its configuration fingerprint is `3e8ab3b57d35190989c47117f139b414b9f656db8a933d6c5d1df6d2aa5fbcc4`. The audit reconstructs panels and estimated BLM firm classes from the saved seeds, but does not refit a successful estimator. Statements borrowed from a paper or software specification are cited where used; otherwise, equations are definitions or explicit derivations.

2 Notation and project estimands

Table 1 gives every symbol needed later. Starting here avoids changing notation as the procedures are introduced.

Table 1: Notation used throughout the experiment

Symbol	Exact definition	Meaning
i, j, t	Worker, firm, and period indices	Unit of the matched panel
Y_{ijt}	$m_{ij} + \varepsilon_{ijt}$	Observed wage outcome
m_{ij}	$\mathbb{E}[Y_{ijt} \mid i, j]$	Complete systematic wage schedule
P_{ij}	Population probability of match (i, j)	Observed assignment law
p_i, q_j	Marginals of P_{ij}	Worker and firm population weights
$p_i q_j$	Product of the two marginals	Independent-assignment counterfactual
a_i, b_j, h_{ij}	Components of $m_{ij} = \mu + a_i + b_j + h_{ij}$ under $p_i q_j$	Worker main effect, firm main effect, and interaction
Q_F	$\mathbb{E}_{pq}[(m_{ij} - \mathbb{E}_q[m_{ij} \mid i])^2]$	Firm-side schedule variation
H_F	$2\mathbb{E}_{pq}[h_{ij}^2]$	Twice the nonadditive interaction variance
ρ_H	$H_F/(2Q_F)$ when $Q_F > 0$	Interaction share of firm-side variation
C_{assign}	$\{\text{Var}_P(m) - \text{Var}_{pq}(m)\}/2$	Contribution of observed assignment to schedule variance
α_i, ψ_j	Pseudo-true additive worker and firm effects	Native KSS/AKM projection objects
λ_i, μ_j	$\mathbb{E}_P[m_{ij} \mid i]$ and $\mathbb{E}_P[m_{ij} \mid j]$	Native BS20 worker and firm wage types

The ANOVA normalization sets the p-weighted mean of a_i, the q-weighted mean of b_j, and both weighted margins of h_{ij} to zero. Consequently,

$$m_{ij} - \mathbb{E}_q[m_{ij} \mid i] = b_j + h_{ij}, \tag{1}$$

$$Q_F = \text{Var}_q(b_j) + \mathbb{E}_{pq}[h_{ij}^2] = \text{Var}_q(b_j) + \frac{H_F}{2}. \tag{2}$$

Equation (2) is a direct derivation from the project definitions, not an assumption about an estimator.

3 Data-generating processes

3.1 Common population and panel design

Every population contains 300 workers and 18 firms. Every panel contains ten periods, hence 3,000 observations before procedure-specific cleaning. The systematic wage schedule has grand mean zero. Worker and firm population marginals are uniform. The observation disturbance is independent and normally distributed with mean zero and standard deviation 0.5.

All Gaussian coordinates are redrawn in every replication. Main-effect vectors are standardized to finite-population mean zero and variance one. Factor columns are centered and orthonormalized under the uniform population weights. Therefore the singular values stated below hold exactly in every realized population, rather than only in expectation.

Observed matching probabilities are defined by

$$P_{ij} = A_i B_j \exp\left\{0.5 g_i f_j + 0.3 \frac{h_{ij}}{\text{sd}(h)}\right\}, \tag{3}$$

where the balancing constants A_i and B_j restore uniform worker and firm marginals. For AKM, $h_{ij} = 0$ and the second sorting term is omitted. A worker's first firm is drawn from the worker-specific conditional distribution. In each later period, the worker redraws from that distribution with probability 0.40 and otherwise stays. A redraw may select the same firm.

3.2 The five wage schedules

Table 2: DGP-specific structure; all omitted common parameters are stated above

DGP	Construction	Nonadditivity
AKM	Independent raw worker and firm $N(0,1)$ effects	Rank zero; $m_{ij} = \alpha_i + \psi_j$ [1]
Crippa/Tukey	Independent raw $\alpha_i, \psi_j \sim N(0,1)$	$h_{ij} = 0.75\alpha_i\psi_j$; rank one; singular value 0.75 [4, Sec. 2.2]
BLM types	Two equal worker types and three equal firm classes; raw type effects and factors are $N(0,1)$	Rank one; singular value 1; genuinely discrete wage surface [2]
Low-rank factors	Worker factor has raw correlation 0.35 with α_i; firm factor has raw correlation 0.25 with ψ_j	Rank two; singular values $(1, 0.5)$
GKLP	Raw $\mathrm{Corr}(Z_i, \eta_i) = 0.35$ and $\mathrm{Corr}(c_j, b_j) = 0.25$	$m_{ij} = Z_i + c_j + b_j\eta_i + \frac{1}{2}b_j^2(0.5)^2$; rank one [5, 6]

The Crippa and GKLP rows adapt the cited wage surfaces to the finite simulated population. The numerical standardization, assignment law in Equation (3), and parameter values in Table 2 are simulation choices, not claims from those papers.

4 Procedures

4.1 KSS

KSS applies leave-out variance-component corrections to a linear model with many effects and unrestricted heteroskedasticity [7]. Here PyTwoWay 0.3.21 supplies the heteroskedastic KSS correction with approximate trace and leverage calculations (`fe_exact=False`) [8]. Its native population schedule is additive:

$$m_{ij}^{\mathrm{KSS}} = \mu + \alpha_i + \psi_j. \tag{4}$$

It natively reports $\mathrm{Var}_q(\psi_j)$ and $\mathrm{Cov}_P(\alpha_i, \psi_j)$.

The additive restriction also implies

$$h_{ij}^{\mathrm{KSS}} = 0, \qquad H_F^{\mathrm{KSS}} = 0, \qquad \rho_H^{\mathrm{KSS}} = 0 \quad \text{if } Q_F^{\mathrm{KSS}} > 0. \tag{5}$$

These are imposed zeros, not interaction quantities estimated from the data. The corrected report includes them to show exactly what additivity implies. KSS residual variance is not relabeled as H_F: under misspecification, that residual mixes observation noise with omitted schedule structure.

4.2 BLM

BLM models a discrete worker type by firm class wage distribution and estimates firm classes before the finite-mixture wage model [2]. Every DGP is fit using two worker types, three estimated firm classes, wage-CDF resolution 10, four initializations, the best two retained starts, at most 250 mover and stayer iterations, and convergence threshold 10^{-6}. The implementation is PyTwoWay 0.3.21 [8].

The actual pipeline is linear:

1. Drop every worker who leaves a firm and later returns to it.
2. Collapse the remaining panel to employment spells.
3. Estimate three firm clusters from wage CDF features using K-means.
4. Convert the spell panel to mover and stayer event-study observations.
5. Require all three stayer firm classes and all nine mover origin–destination class pairs.
6. Fit four mover-model starts and retain the two best likelihoods.
7. Select among those starts using PyTwoWay's connectedness calculation.
8. Fit the stayer model conditional on the selected mover model.

Step 5 is the audit correction. PyTwoWay 0.3.21 reshapes probability counts to 2×3^2 for movers and 2×3 for stayers without padding absent categories. Without a preflight support check, a missing terminal label raises a reshape exception, while a missing interior label can instead create a zero probability row. The corrected code detects the common economic cause before calling the optimizer.

For continuous DGPs, BLM receives no oracle worker or firm labels. Deterministic equal-count groups based on product-marginal mean wages are used only to score the estimated cell table. Its cell-table functionals are also compared with the full project schedule, so discretization error remains visible.

4.3 BS20

BS20 estimates moments of worker and firm wage types in matched data [3]. PyTwoWay's weighted standard estimator is used after strongly-connected, spell-level, no-return cleaning [8]. The native population covariance is

$$C_{\mathrm{BS}} = \mathrm{Cov}_P(\lambda_i, \mu_j), \qquad \lambda_i = \mathbb{E}_P[m_{ij} \mid i], \quad \mu_j = \mathbb{E}_P[m_{ij} \mid j]. \tag{6}$$

The code reads PyTwoWay's `cov(lambda, mu)` directly. It does not construct that number from project quantities. Because C_{BS} is not generally C_{assign}, BS20 is no longer displayed as an estimator of C_{assign}. Their population difference is reported separately as an estimand comparison.

4.4 Project low-rank plug-in

The current project fit is

$$m_{ij} = \alpha_i + \psi_j + \sum_{\ell=1}^{r} s_\ell U_{i\ell} V_{j\ell}. \tag{7}$$

Positive ranks use three starts, tolerance 10^{-6}, and at most 300 iterations. Candidate ranks are $\{0, 1\}$ for AKM and grouped BLM and $\{0, 1, 2\}$ for Crippa, continuous low-rank, and GKLP. All candidates use a common retained core: minimum degree 3 for $\{0, 1\}$ and degree 4 for $\{0, 1, 2\}$.

The selected rank minimizes

$$\mathrm{BIC}(r) = n \log(\mathrm{SSE}_r / n) + \mathrm{df}(r) \log n, \qquad \mathrm{df}(r) = N + J - 1 + r(N + J - 2 - r). \tag{8}$$

Equation (8) is an exploratory simulation rule. It is not part of the unfinished project LOO theory.

5 Evaluation rules

The experiment separates four statuses:

- **Pass/completed.** KSS and BS20 returned a value. BLM additionally has complete support and no one-step likelihood decline greater than 10^{-4}. The project fit satisfies convergence and multi-start diagnostics.
- **Returned with warning.** An admissible BLM or project fit returned finite values but failed its procedure-specific numerical diagnostic.
- **Unsupported.** BLM lacks a stayer class or mover class-pair after its own cleaning and clustering. It is not sent to the optimizer.
- **Estimator failure.** An admissible sample reaches the estimator, but the estimator returns no value.

The 10^{-4} BLM likelihood threshold is an engineering diagnostic, not a statistical critical value. For a positive-rank project fit, at least two of three starts must have objectives within $0.0001 \max(1, |L^*|)$ of the best objective, and their Q_F, H_F, and C_{assign} spreads must each be within $0.001 \max(1, |\theta^*|)$. These thresholds are also engineering checks. A pass does not prove correct specification, correct rank, or low bias.

RMSE uses all values returned from admissible samples. Unsupported samples do not enter the RMSE. A pass-only RMSE is reported separately and is not treated as unconditional procedure performance.

6 Results

6.1 Feasibility and numerical status

Table 3 and Figure 1 report pass/warning/unsupported/failure counts. For KSS and BS20, "pass" means only that PyTwoWay completed.

Table 3: Status counts out of 100 replications: pass / warning / unsupported / failure

DGP	KSS	BLM	BS20	Project
AKM	100/0/0/0	43/0/57/0	100/0/0/0	100/0/0/0
Crippa/Tukey	100/0/0/0	57/0/43/0	100/0/0/0	100/0/0/0
BLM types	100/0/0/0	46/1/53/0	100/0/0/0	6/94/0/0
Low-rank factors	100/0/0/0	46/0/54/0	100/0/0/0	78/22/0/0
GKLP	100/0/0/0	50/0/50/0	100/0/0/0	94/6/0/0

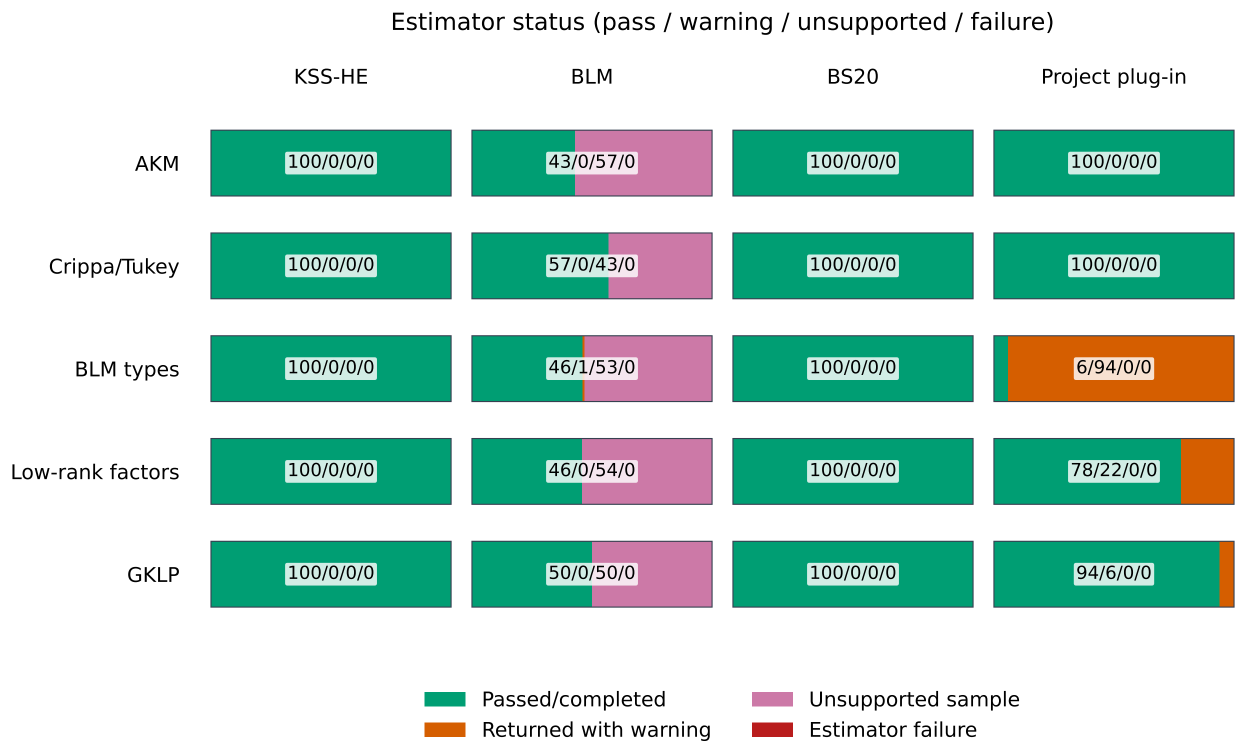

Figure 1: Estimator status. The four numbers printed in each cell follow the order in Table 3.

The BLM audit finds 257 unsupported samples and zero estimator failures. The unsupported samples consist of 3 with no stayers, 246 with at least one absent stayer class, 6 with only a missing mover class-pair, and 2 with both a missing stayer class and mover class-pair. Thus 243 samples have complete support; 242 pass and one returns with a likelihood-path warning.

The high unsupported rate follows from the DGP rather than random-start failure. As an approximation, the probability of no actual firm change in a later period is

$$0.6 + \frac{0.4}{18} = 0.622. \tag{9}$$

The probability of remaining at the same firm through nine later periods is $0.622^9 = 0.014$, or about four workers out of 300. After returners are dropped, those few stayers must cover three firm classes. The approximation explains the scarcity; the reported classification uses exact realized support.

6.2 Correctly specified benchmarks

The expected own-model benchmarks work. Under AKM, KSS-HE's bias for Q_F is $+0.0032$ and its RMSE is 0.0485; its bias for C_{assign} is -0.00014. Under the grouped BLM DGP, the 47 admissible returned BLM fits have the results in Table 4. These are conditional on support and therefore do not erase the 53 percent infeasibility rate.

Table 4: BLM under its grouped DGP, among 47 admissible returned fits

Project functional	Bias	RMSE
Q_F	-0.0054	0.0492
H_F	-0.0031	0.0741
ρ_H	$+0.00056$	0.0140
C_{assign}	$+0.0025$	0.0110

6.3 Common project targets

Figure 2 compares admissible returned outputs with the common project truth. KSS is shown for H_F and ρ_H because Equation (5) implies structural zeros. The comparison therefore measures what the additive restriction misses; it is not an estimated KSS interaction. BS20 is absent from the C_{assign} panel because Equation (6) defines a different covariance.

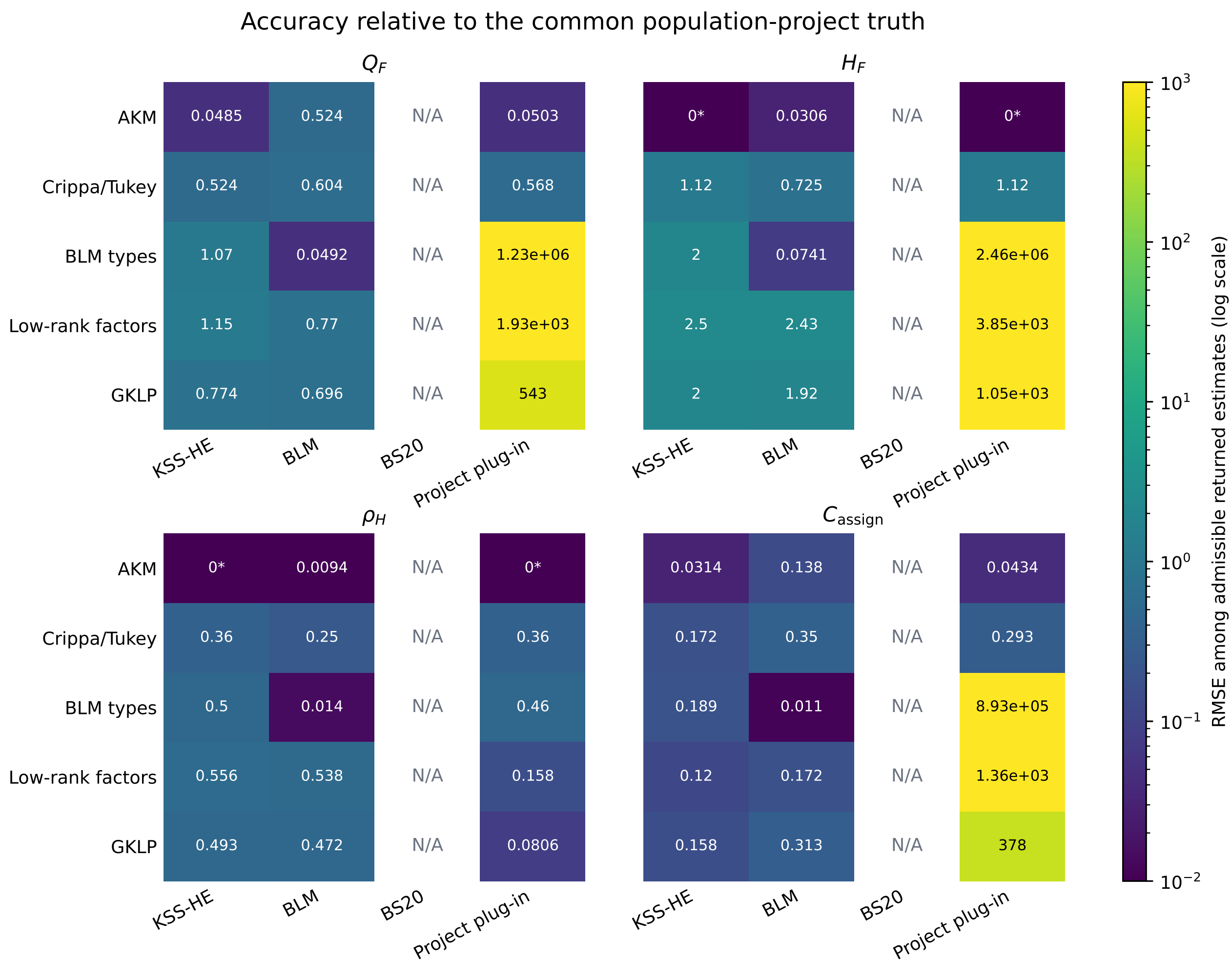

Figure 2: RMSE relative to the common project truth among admissible returned estimates. N/A means there is no defensible mapping to that project object. The label 0* denotes an AKM structural zero.

Three patterns matter. First, KSS's H_F and ρ_H errors under the nonadditive DGPs are exactly the magnitudes excluded by additivity. Second, supported BLM fits recover the grouped DGP but lose substantial interaction variation when a continuous schedule is compressed into six cells. Its H_F RMSE is 0.725 under Crippa, 2.433 under continuous low rank, and 1.923 under GKLP. Third, the project estimator's all-returned RMSE can be dominated by a small number of positive-rank warning fits. Figure 5 separates this tail risk from its pass-only behavior.

6.4 Native targets versus project targets

For procedure p, comparison with the project truth decomposes as

$$\widehat{\theta}_p - \theta_{\text{proj}} = \underbrace{(\widehat{\theta}_p - \theta_p)}_{\text{estimation error}} + \underbrace{(\theta_p - \theta_{\text{proj}})}_{\text{estimand difference}} . \tag{10}$$

Figure 3 and Table 5 report the second term averaged over all 100 populations. They contain no estimator sampling error.

Table 5: Mean native population target minus project population target

DGP	KSS firm variance $-Q_F$	KSS covariance $-C_{\text{assign}}$	$C_{\text{BS}} - C_{\text{assign}}$
AKM	+0.000	+0.000	+0.416
Crippa/Tukey	−0.394	−0.0028	+0.824
BLM types	−0.752	+0.014	+0.622
Low-rank factors	−1.127	−0.017	+0.459
GKLP	−0.741	−0.080	+0.452

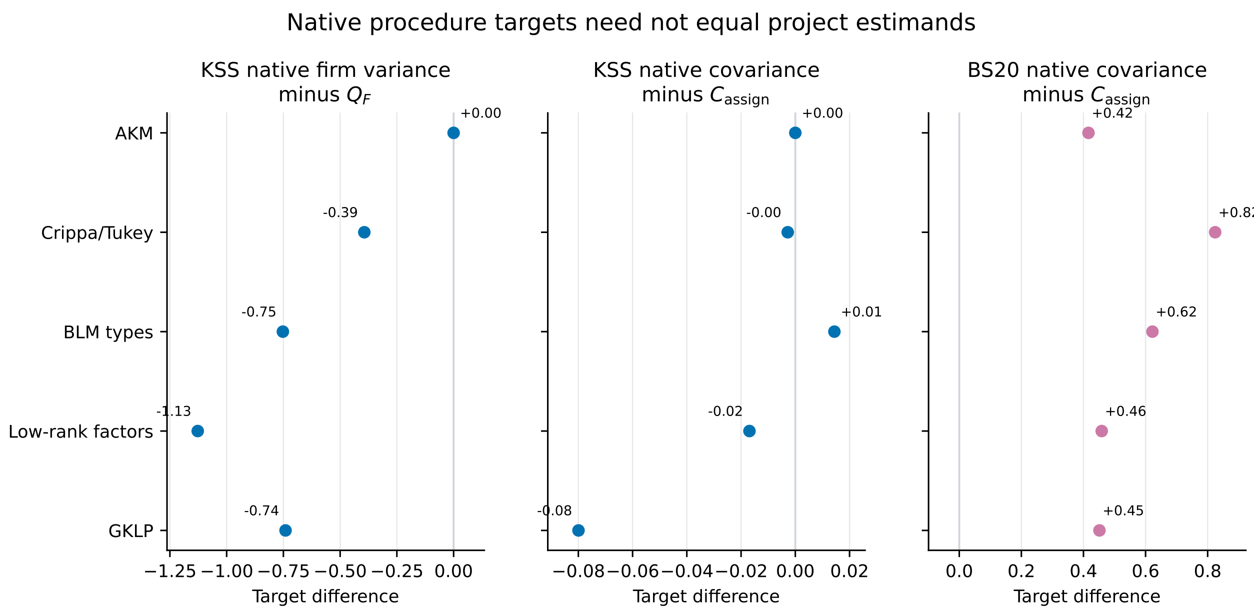

Figure 3: Population estimand differences. These points are native targets minus project targets, not estimator bias.

The negative KSS firm-variance differences have an exact explanation under independent assignment.

The additive projection then recovers b_j, so Equation (2) gives

$$\operatorname{Var}_q(\psi_j) - Q_F = -\frac{H_F}{2} \le 0. \tag{11}$$

With sorted assignment, the pseudo-true additive firm effect can absorb part of h_{ij}, so Equation (11) no longer fixes the sign. The negative values in the four sorted nonadditive designs are therefore informative simulation results, not a universal theorem. The KSS covariance gaps are small and of mixed sign, confirming that no analogous sign claim is warranted.

The BS20 object also needs its own explanation. Under an additive schedule $m_{ij} = a_i + b_j$, define the conditional-expectation operators $Tb(i) = \mathbb{E}[b_J \mid I = i]$ and $T^*a(j) = \mathbb{E}[a_I \mid J = j]$. Then

$$\lambda_i = a_i + Tb(i), \qquad \mu_j = b_j + T^*a(j), \tag{12}$$

$$C_{\text{BS}} - C_{\text{assign}} = \|Tb\|^2 + \|T^*a\|^2 + \langle Tb, TT^*a\rangle \ge 0. \tag{13}$$

The inequality follows from Cauchy–Schwarz and the fact that conditional expectation is a contraction in mean square. Thus the positive AKM gap has a mathematical explanation: sorting enters both conditional wage types. For a general nonadditive schedule, Equation (13) need not hold. The positive gaps in the other four rows are features of these DGPs, not a general property imposed by the code.

BLM has another population difference outside its discrete DGP. Replacing a continuous wage surface with a 2×3 grouped surface changes the project functionals even before estimation. The grouped-minus-full H_F gaps are -0.591 for Crippa, -2.452 for continuous low rank, and -1.959 for GKLP. The grouped BLM DGP has zero gap up to floating-point error because its true labels already define the scoring cells.

6.5 Rank selection

Rank is the number of interaction dimensions in Equation (7). Rank zero is additive; rank one permits one comparative-advantage dimension; rank two permits two. Figure 4 shows how often BIC selects each candidate. The orange outline marks the true rank, while the text at right is the separate multi-start diagnostic-pass rate.

Table 6: Project BIC rank selection

DGP	True rank	Select 0	Select 1	Select 2	Pass rate
AKM	0	100	0	0	100%
Crippa/Tukey	1	100	0	0	100%
BLM types	1	2	98	0	6%
Low-rank factors	2	0	100	0	78%
GKLP	1	0	100	0	94%

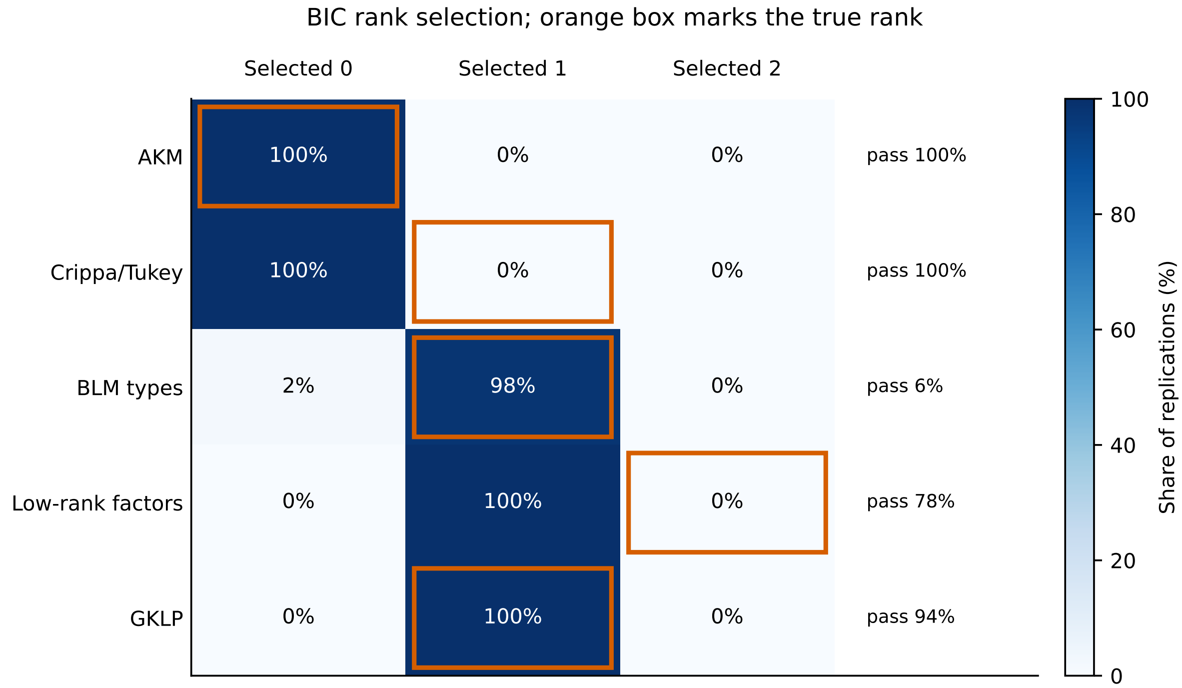

Figure 4: Share of replications selecting each rank. Rank accuracy and the numerical pass rate answer different questions.

Crippa makes the distinction sharp: every project fit passes numerically, but BIC selects rank zero in every replication. The estimator therefore sets the true rank-one interaction to zero. In the rank-two DGP, BIC always selects rank one, consistent with retaining the stronger singular-value-1 direction and discarding the weaker singular-value-0.5 direction. A future LOO bias correction cannot repair an interaction dimension removed by rank selection.

6.6 Warnings and tail risk

Figure 5 compares project RMSE among all returned fits with RMSE conditional on passing the numerical diagnostics. A large gap shows that a small warning subset dominates squared error. The pass-only line is useful for diagnosis, but it conditions on an estimation outcome and cannot replace unconditional performance.

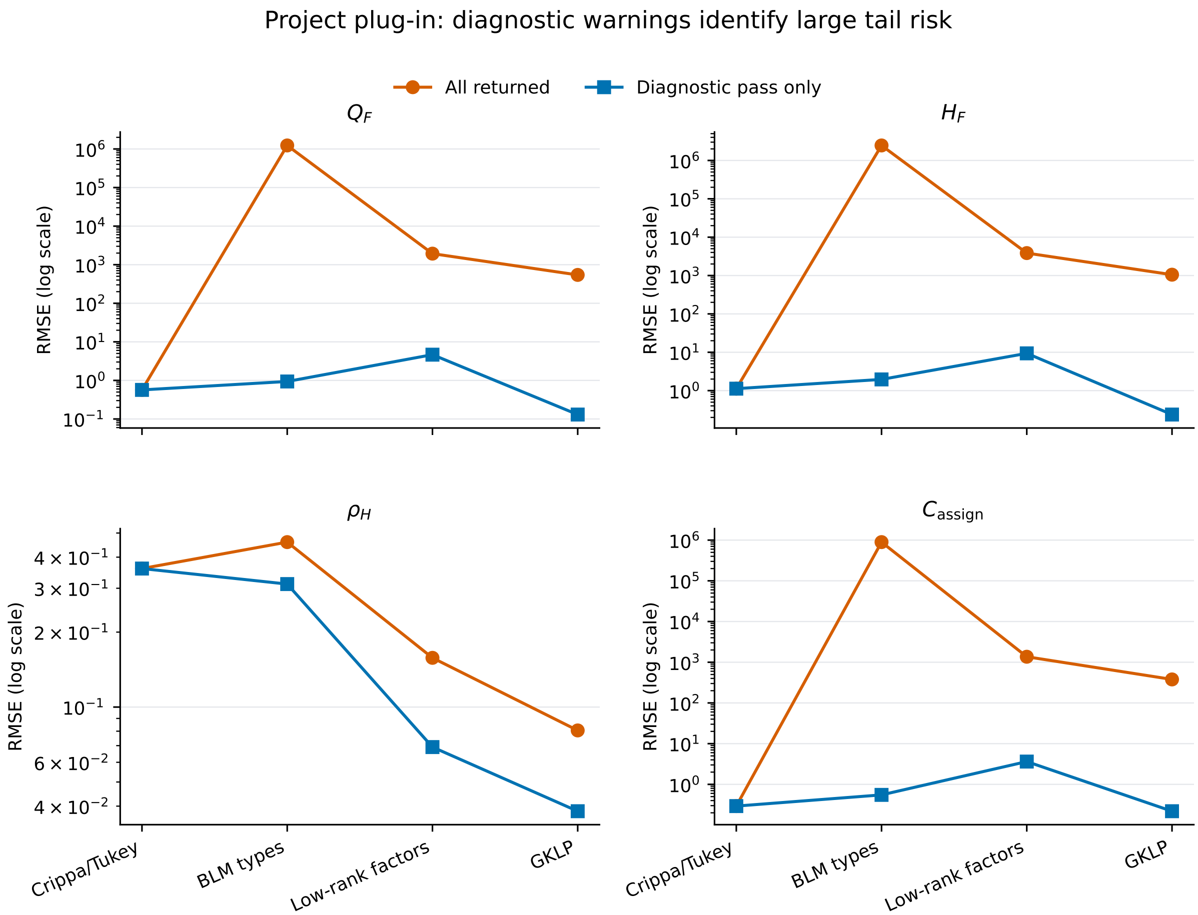

Figure 5: Project plug-in RMSE among all returned fits and conditional on a diagnostic pass. The vertical axis is logarithmic.

7 Interpretation and next simulation changes

The full DGP-by-estimator matrix answers the original question more clearly than an all-AKM exercise. KSS works in its additive benchmark, and BLM works in its discrete benchmark when its support requirements are met. Outside those benchmarks, performance changes for two different reasons: estimators can be inaccurate for their native target, and native targets can differ from the project target. Equation (10) shows why those sources must not be combined without explanation.

The corrected BLM result is not "the optimizer fails half the time." The current observation design produces too few stayers after no-return cleaning. More than half of the panels therefore do not identify every probability row that the configured 2×3 BLM model asks PyTwoWay to update. A fair next experiment should retain the present design as a support stress test and add a BLM-feasible design—for example, more workers, a permanent-stayer mixture, a lower redraw probability, or fewer firm classes. That is a DGP change and should be declared in advance rather than hidden inside the estimator.

For the project procedure, the dominant present issue is rank selection and positive-rank reproducibility, not only small-sample plug-in bias. Crippa shows complete under-selection with numerical convergence; the rank-two design shows systematic truncation; and the warning fits create very large squared-error tails. Rank choice and stable schedule completion therefore need to be solved before a leave-out correction can be interpreted as the principal remaining improvement.

Finally, the reporting distinctions are substantive. KSS's interaction zeros are now labeled "imposed," BS20 is reported under its native covariance rather than under C_{assign}, and BLM support failures are separate from optimizer failures. These labels prevent a numerical table from silently changing the meaning of an estimand.

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